

Unit 12, Lesson 1: Applying the Distributive Property with Variables

Benchmark

MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.

Learning Target

- ☐ I can write equivalent algebraic expressions by applying the distributive property.

12.1.1 Warm-Up: Growth Mindset

1. On a scale from 1 to 5, rate each of the following statements by circling the number that matches your belief.

1 = strongly
disagree

2 = disagree

3 = neither/
neutral

4 = agree

5 = strongly
agree

a. Intelligence is something you're born with.	1	2	3	4	5
b. I can get smarter if I work hard.	1	2	3	4	5
c. Some things are just too difficult for some people to learn.	1	2	3	4	5

2. Which statement did you agree with most? _____

3. Explain why you agree with it most.

12.1.2 Refresher: Multiplying with Exponents

Any variable, a , can be rewritten using the exponent 1, as a^1 . This is the identity exponent law.

1. Complete the table by applying the identity exponent law and product of powers law to each given expression. The first row has been started.

Expression	Equivalent Expression Using Identity Exponent Law	Equivalent Expression Using Product of Powers Law
$d \cdot d$	$d^{\square} d^{\square}$	$d^{\square}$
$(4x)(x)$		
$(-3f)(0.5f)$		
$5c \cdot b \cdot 2c$		

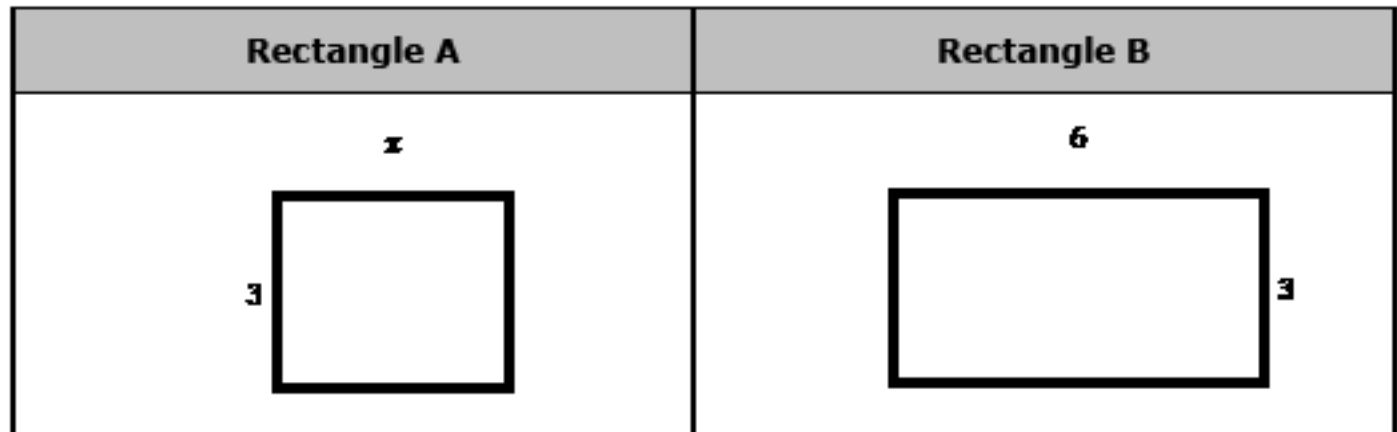
2. Complete the statement.

Any variable, a , multiplied by itself can be rewritten as $a^{\square}$.

12.1.3 Exploration: Areas of Rectangles

Work with your partner to complete the following.

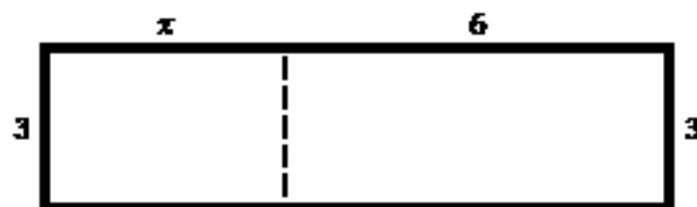
1. Two rectangles are shown. The length and width of Rectangle B are 6 and 3. The width of Rectangle A is also 3, but its length is unknown, so it is labeled x .



- a. Write an expression to represent the sum of the areas of Rectangles A and B.

_____ + _____

- b. Rectangles A and B can be composed into one larger rectangle, as shown.



Complete the statement.

The width of the new rectangle is

- ☐ 3
☐ $3x$
☐ 6
☐ 18

and the length is

- ☐ 3.
☐ $(3 + x)$.
☐ 6.
☐ $(x + 6)$.

- c. Tajae and Elyse wrote different expressions to represent the area of the larger rectangle.

Tajae	Elyse
$3(x) + 3(6)$	$3(x + 6)$

Discuss with your partner whether Tajae, Elyse, or both students wrote a correct expression of the area.

- d. Write another expression to represent the area of the larger rectangle.

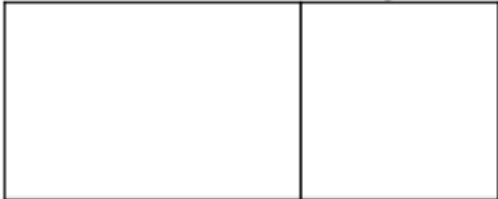
2. For each rectangle, complete the table.

In the first two columns, write an expression to represent the width and length of the rectangle. Then, in the next two columns, express the area of the rectangle in two different ways using either multiplication or addition.

Rectangle A		6 x $\frac{2}{3}$	
Width	Length	Area (width x length)	Equivalent Area Expression
		___ (___)	___ + ___

Rectangle B	$4f$ 5 2 		
	Width	Length	Area (width x length)

Rectangle C	4.2 6 d 		
	Width	Length	Area (width x length)

Rectangle D	7 y y 		
	Width	Length	Area (width x length)

12.1.4 Bring It Together: Distributive Property

The area models used throughout the Exploration demonstrate the **distributive property of multiplication over addition**.



Distributive Property of Multiplication Over Addition

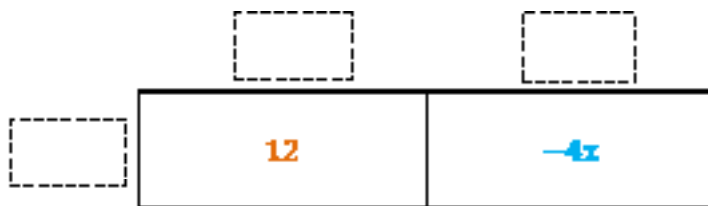
$$a(b + c) = (a \times b) + (a \times c)$$

The factor outside of parentheses is “distributed” to each part of the factor within a set of parentheses. With the distributive property, multiplication is performed over (before) addition or subtraction.

1. Discuss with your partner how multiplication being performed before addition or subtraction relates to the order of operations.
2. Consider the example that shows the application of the distributive property to multiply.

$$4(3 - x) = 12 - 4x$$

- a. Complete the area model to represent the same expressions by filling in the blanks for each factor.



- b. Discuss with your partner how the area model shows the distributive property.

- c. In the product $12 - 4x$, explain why the constant term is positive and the variable term is negative.
- d. Zion says the product could also be written as $-4x + 12$. Explain whether or not this is true.

3. The distributive property can also be used to multiply algebraic expressions where variables are included in both factors. An example is shown.

$$x(3 - x) = 3x - x^2$$

Complete the statements.

The product of $x \cdot 3$ is ☐ $3x$.
☐ $-x^2$. The exponent of x in this term is ☐ 1
☐ 2 as

explained by the ☐ identity exponent.
☐ product of powers.

The product of $x \cdot (-x)$ is ☐ $3x$.
☐ $-x^2$. The exponent of this term is ☐ 1
☐ 2 as

explained by the ☐ identity exponent.
☐ product of powers.

Therefore, the ☐ factors
☐ product of $x(3 - x)$ is $3x - x^2$, which is determined by

applying the distributive property to multiply the ☐ factors
☐ product x and $(3 - x)$.

12.1.5 Guided Practice: Distributing Terms with Variables

1. Generate an equivalent algebraic expression for each given expression by applying the distributive property.

a. $f(f - 6)$

b. $(0.2g + 1)5g$

c. $\frac{3}{2}c\left(2c + \frac{3}{2}\right)$

d. $-4x(x - 7)$

2. What is another equivalent expression for $-4x(x - 7)$ using the commutative property?

12.1.6 Wrap-Up: Ticket Out the Door

Rewrite each expression using the distributive property.

1. $(8 + 4.5z)(0.25x)$

2. $-15y\left(\frac{2}{5}y - \frac{1}{2}\right)$